

Can the variability of trend-following signals add value?

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Abstract

We document that there is information in the variability of binary signals used to build a cross-asset trend-following strategy. This information may help building trend-following strategies with slightly higher Sharpe ratios. This added value may come with higher maximum drawdown to vol ratios for short lookback periods (up to one month), while the longest period tested (2.5 months) lead to a reduction of both ratios. The optimal results are obtained for observation periods of 1 to 2 months.

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1 Introduction

The famous quote “It is difficult to make predictions, especially about the future” has been ascribed to many people over the past 70 years (it seems it started becoming popular in the mid 1950s), from Mark Twain to the Nobel prize-winning physicist Niels Bohr. In financial markets, if the efficient market hypothesis was true, then making predictions using public data would be a fool’s errand: statistical arbitrage would be impossible, and trend-following would not work very well. In practice though, some statistical arbitrageurs and trend-followers are successful. Even if a large literature using either information theory (mainly based on Shannon’s entropy) and fractal analysis (mainly based the Hurst exponent) addresses the feasibility of forecasting, it remains difficult to time arbitrage and trend-following strategies, and it remains difficult to time the usefulness of the individual assets in a trend-following strategy.

In this paper, we explore the alteration of simple trend-following signals based on their own time-series variability, using variability functions derived from machine learning (which have relied, among other functions, on Shannon’s entropy). Hence, we do not study the relevance of the Hurst exponent, which we leave for a latter paper. We find empirically that taking variability into account can indeed add value to a trend-following strategy, and that there is a sweet spot for the lookback period used to compute variability of signals.

The rest of the paper is organised as follows. Section 2 provides an overview of our dataset. Section 3 describes the construction of trend-following strategies, and Section 4 details how variability is defined and how the original signals are transformed. Section 5 evaluates empirically the trend-following strategies, and finally, Section 6 concludes.

2 Data Description

In this section, we describe briefly the various datasets that we use in this paper. All data are daily, and when not mentioned, are from January 1999 to June 2024. The global dataset comprises 4 main asset classes: Fixed Income, FX, Equities and Commodities. Fixed Income, Equities and Commodities data are obtained by rolling futures contracts with Bloomberg (the first contract is used then rolled into the next one a few days before expiry). For FX, 3-month forwards are used.

The investment universe of Fixed Income consists of 15 futures contracts across Australia, Canada, France, Germany, Italy, Japan (to January 2017), South Korea (from January 2017) , the UK and the US. For all countries, the generic 10y contract is used. For Germany and the US, the 2y, 5y and 30y contracts are also included.

The universe of 3-month currency forwards covers the following exchange rates:

- all 9 G10 currencies vs USD: AUD (Australia), CAD (Canada), EUR (Eurozone), JPY (Japan), NZD (New Zealand), NOK (Norway), SEK (Sweden), CHF (Switzerland) and GBP (UK)
- 13 non G10 currencies vs USD: BRL (Brazil), CLP (Chile), COP (Colombia, from 2017), CZK (Czech Republic, from February 2022), HUF (Hungary), INR (India), IDR (Indonesia), KRW (South Korea), MXN (Mexico), PLN (Poland), SGD (Singapore), TWD (Taiwan), ZAR (South Africa)

The investment universe of equities futures consists of:

- 18 indices of G10 countries: ASX (Australia), TSX (Canada), CAC (France), DAX (Germany), EuroStoxx (Euro Area), FTSE/MIB (Italy), TOPIX and Nikkei (Japan), IBEX (Spain), OMX (Sweden), SMI (Switzerland), FTSE (United Kingdom), S&P 500, Dow Jones, Nasdaq 100, Russell 2000 (from October 2017) and S&P 400 (United States).
- 11 indices of non G10 countries: Bovespa (Brazil), Hang Seng (China and Hong Kong), NSE (India), Kospi (South Korea), FTSE KLCI (Malaysia), Mexican Bolsa (Mexico), WIG (Poland), FTSE (Taiwan), SET (Thailand), FTSE/JSE (South Africa)

The commodities investment universe consists of the 24 commodities that are in the BCOM as of end of June 2024. Constituents of the BCOM may move from one year to the other: for instance Low Sulphur Gas Oil was added in 2019, Cocoa was removed in 2005 (but considered for inclusion in 2024), hence resetting the universe to the BCOM constituents would be more realistic than this static universe. However the commodities currently in the BCOM have been around for quite some time, and most have been liquid enough for the history used of this paper, except for Copper, Soybean Oil and Soybean Meal (from 2010).

Our investment universe therefore comprises 90 instruments, with over 75 from day one.

3 Portfolio Construction

3.1 Original signal

The toy model used in this paper relies on simple signals. For a given lookback period, the signal of an asset is the sign of its cumulative return: +1 (positive) or -1 (negative). More formally, the signal is, at date t , for an asset k , and a lookback period p :

$$s_{t,p}^k = \text{sign}\left(r_{t-p,t}^k\right)$$

where $r_{t-p,t}^k$ represents the transformed cumulative returns for asset k from date $t - p$ to date t .

Literature shows trend-following is strongest using a 3 to 12-month lookback period, and also exists for smaller periods such as one month, though there is a trade-off between signal and trading costs for short lookback periods. To build a robust signal, we retain four lookback periods: 1.5, 3, 6 and 12 months (for a quick justification of why the following lookback period is twice the previous lookback period, see [7]).

The aggregate signal for a given asset k is therefore:

$$s_t^k = \frac{1}{4} \sum_{l=1}^4 s_{t,l}^k$$

with $l \in [1.5; 3; 6; 12]$ months.

3.2 From signal to portfolio

Qian ([14] and [15]) and Maillard, Roncalli and Teiletche ([11]) popularized the risk parity approach in portfolio construction in a long-only framework. Their work was soon extended to the risk budgeting approach (see for instance [5]), in both long-only and long-short framework. Baltas ([1]) and Harvey et al. ([8]) document the interest of using both volatility targeting and a risk budgeting framework in a multi-asset trend-following strategy. We follow a similar approach, by building a multi-asset long-short portfolio per lookback period, with our four lookback periods ranging from 1.5 to 12 months.

Hence, we will take a long (resp. short) position in asset k if it has performed positively (resp. negatively) over three or four periods considered, while the position will be neutral if it has performed positively and negatively for two lookback periods. Given this definition, the odds of a signal with a value of 0 are not negligible. When the signal for a given asset is 0, its risk budget will not be allocated to other assets.

We chose to split the risk budget evenly between the four asset classes (Fixed Income, FX, Equities and Commodities), ie 25%. Within these asset classes, risk budget allocations are as follows:

- Fixed Income: equal risk contribution. There are four German and US assets in the investment universe, hence an equal risk framework acknowledges the major influence of these assets.
- FX: risk equally split between G10 and non G10 currencies, and equal risk contribution within these two buckets. This helps harvest diversification benefits from non G10

currencies, while making sure the overall FX risk budget remains balanced.

- **Equities:** in between an equal risk contribution and market cap weighted risk contribution. Like for FX, this strikes an interesting balance between liquidity and diversification.
- **Commodities.** We choose to allocate specific risk budgets for the sectors of the BCOM: Energy, Industrial Metals, Precious Metals, Grains, Softs and Livestock (Energy being allocated the largest budget, Livestock the smallest), and within each sector, risk budgets are uniform.

At date t , the vector of portfolio weights w_t is the solution of the following optimisation problem (see Eq. 21 in [1]):

$$\text{Max} \sum_{k=1}^{N_t} \log(|w_t^k|) \quad (1)$$

subject to

$$w_t' \Sigma_t w_t \leq \sigma_{target}$$

with:

- N_t the number of assets. Though not constant, it is not very volatile, as assets are not frequently added or removed
- Σ_t the covariance matrix of all assets
- σ_{target} the targeted volatility of the risk-budgeted portfolio, which is 10% in this paper

We choose to use the simplest estimate of the covariance matrix Σ_t : the sample covariance matrix, annualized, and calculated over 6 months. 6 months is close to the average lookback period of 5.625, and results using this figure are qualitatively equivalent. To ensure no look-ahead bias, the signals and covariance matrix used at date t are calculated with data up to date $t - 1$.

Signals are calculated on a daily basis, and portfolios are rebalanced on a daily basis as well, at market close. Finally, all returns are expressed in excess of cash, net of expected transaction costs, but gross of management fees.

4 Altering the signal

4.1 Underlying idea

The main idea of this paper is to test whether the variability (or lack thereof) of the aggregate signal has information, by altering the final signal used for portfolio construction

according to the variability. One straightforward way to calculate the variability of the aggregate signal is to use a standard deviation. One would then need to normalize this value so as to use it easily in the portfolio construction process.

Rather than calculating a typical financial volatility, we turn to machine learning, and more particularly to decision trees.

4.2 Decision tree

A decision tree is a hierarchically organized structure, which, at each node, divides data according to the value of a feature. A tree splits the original dataset into p separated subsets. The *parent* of a node c is its immediate predecessor. The *children* of node c are its immediate successors (they have c as their parent). *Leaves* (or leaf nodes) are nodes without children.

A well-know example of a decision tree can be found in the scikit-learn documentation (cf Scikit-Learn documentation).

Let's consider an initial dataset $\mathcal{D} = (x_i, y_i)_{1 \leq i \leq n}$, where features x are used to explain y , a qualitative variable. The goal of a classification tree is to minimize the error (or uncertainty) in each leaf. For any subset R_j , the probability for an observation to be in class k is:

$$\hat{\beta}_{jk} = \frac{\sum_i \mathcal{I}(y_i = k) \mathcal{I}(x_i \in R_j)}{\sum_i \mathcal{I}(x_i \in R_j)}$$

This is the frequency with which label k appears in subset R_j . The obvious question is: how to minimize the error?

A split in a tree occurs only and only if it helps minimizing the error, ie the information gain induced by creating children is always positive. Let's consider the initial dataset $\mathcal{D} = (x_i, y_i)_{1 \leq i \leq n}$, d the number of available features, and focus on a binary tree (ie a tree where nodes have either 2 children, or none).

For all available features, ie $j \in [1, d]$, for each value v_{jk} that can be used as a threshold, the tree algorithm:

- Splits the dataset between $I_- = \{i : x_{ij} < v_{jk}\}$ and $I_+ = \{i : x_{ij} \geq v_{jk}\}$
- Estimates $\beta_- = \frac{\sum_i \mathcal{I}(y_i=1) \mathcal{I}(x_i \in I_-)}{\sum_i \mathcal{I}(x_i \in I_-)}$ and $\beta_+ = \frac{\sum_i \mathcal{I}(y_i=1) \mathcal{I}(x_i \in I_+)}{\sum_i \mathcal{I}(x_i \in I_+)}$
- The quality of the split is measured by the weighted sum of uncertainties:

$$\frac{|I_-|}{|I_-| + |I_+|} H(I_-) + \frac{|I_+|}{|I_-| + |I_+|} H(I_+)$$

The tree algorithm then selects the split that minimizes this weighted sum of uncertainties. This is repeated for each child until the stop criterion is met (for instance a minimum number

of observations in each leaf or the length of the tree). The weighted sum of uncertainties is the conditional uncertainty at a given node C . In other words, information gain is the difference between the parent's information and the information of its children:

$$IG(\mathcal{C}, Children) = H(\mathcal{C}) - H(\mathcal{C}|Children)$$

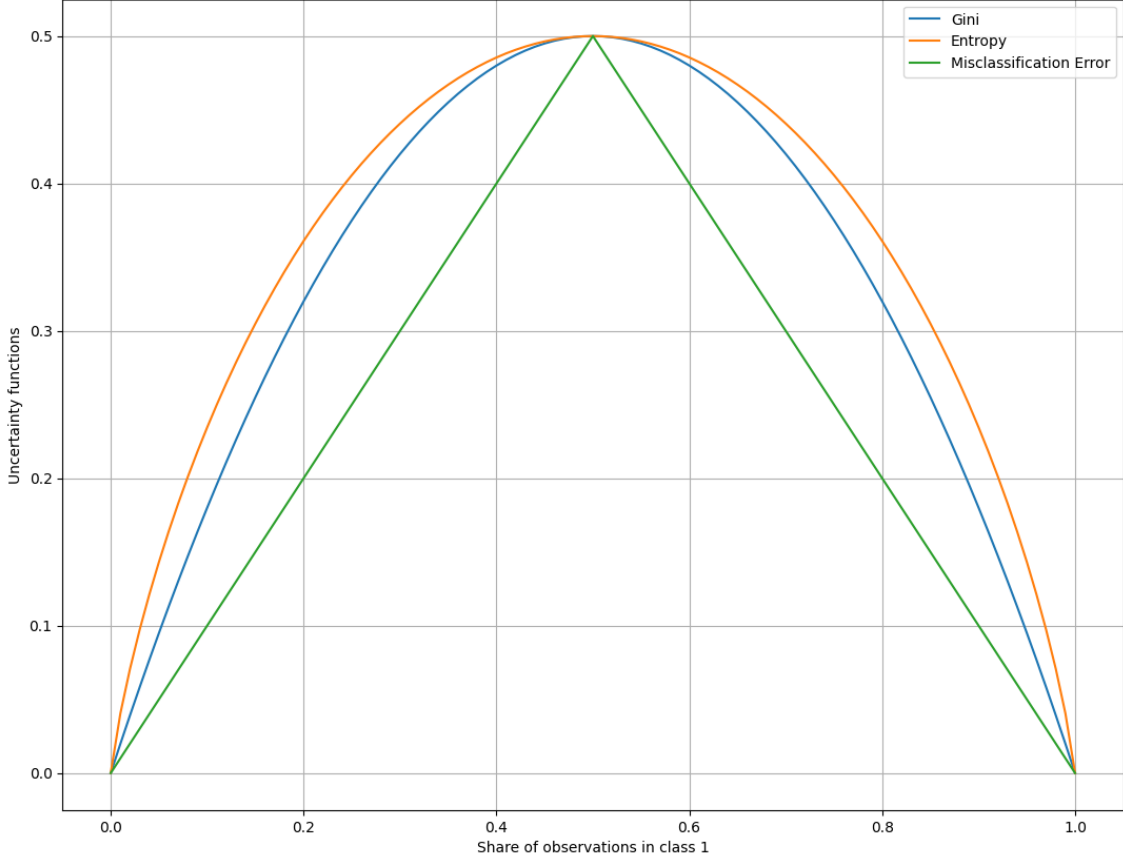
By choosing the split that minimizes the weighted sum, the information gain is maximized.

Uncertainty H can be measured in a variety of ways, for instance, considering X a discrete random variable with K values $(v_1, \dots, v_K)$:

- Entropy: $H(X) = -\sum_{k=1}^K \mathcal{P}(X = v_k) \log(\mathcal{P}(X = v_k))$
- Gini Index: $H(X) = \sum_{k=1}^K 2\mathcal{P}(X = v_k)(1 - \mathcal{P}(X = v_k))$
- Misclassification error: $H(X) = 1 - \max\{\mathcal{P}(X = v_k)\}$

Figure 1 displays these three usual uncertainty functions for a 2-class problem. For the remainder of this paper, we use the Gini Index as our base measure of uncertainty. Results using Entropy are qualitatively equivalent.

Figure 1: Uncertainty functions in class 2 for a 2-class problem, based on p , the share of observations in class 1



4.3 From uncertainty to signal alteration

We remind that the aggregate signal for a given asset k is $s_t^k = \frac{1}{4} \sum_{l=1}^4 s_{t,l}^k$, with $l \in [1.5; 3; 6; 12]$ months. At any time t , for any asset k , $s_t^k \in \{-1; -\frac{1}{2}; 0; \frac{1}{2}; 1\}$, ie classifying s_t^k is a 5-class problem. Over a given lookback period p , ie from $t - p + 1$ to t , we note $\mathcal{P}_l$ the share of time s^k spent in value l . The uncertainty measure is therefore:

$$H(X) = \sum_{l=1}^5 2\mathcal{P}_l(1 - \mathcal{P}_l)$$

H has a maximum value when all states are equally probable, ie when $\mathcal{P}_l = \frac{1}{5}$. We choose to divide H by $2(1 - \frac{1}{5})$ so that it takes values in $[0; 1]$.

Figure 2: Cumulative Distribution Functions of average uncertainties based on the Gini Index, across all assets, with different lookback periods

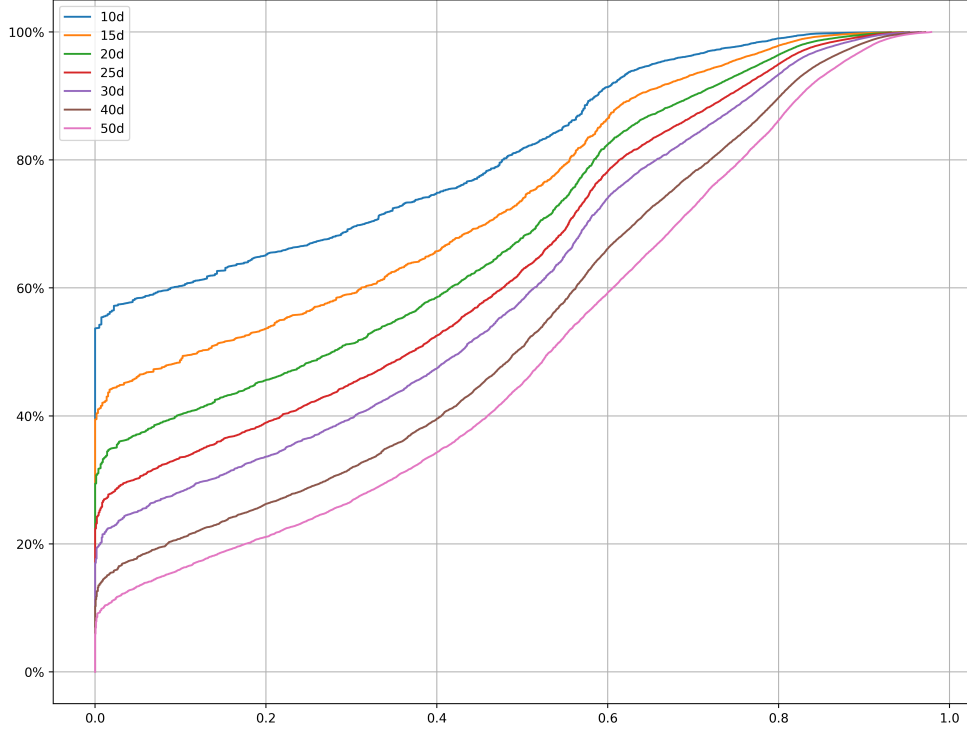


Figure 2 displays the cumulative distribution functions of average uncertainties across all assets, using seven lookback periods ranging from 10 to 50 days. Using several lookback periods is mandatory in order to test the robustness of the approach. Intuitively, the shorter the lookback period, the higher the odds of certainty. This is confirmed empirically: with a 10-day lookback period, roughly 54% of the observations show no uncertainty (ie the signal is unchanged), while this figure drops to 17% with a 30-day lookback period. It is quite clear that the distribution of uncertainties is on average skewed to the left.

We want to penalize the signal if uncertainty has been high, and/or strengthen its value if uncertainty has been low. To achieve this, we transform the uncertainty value into an alteration coefficient c , as displayed in Figure 3, and the final signal s^* will be a combination of the initial aggregate signal s and c : $s^* = s \times c$. Based on this alteration and the skewness of the CDFs in Figure 2, it becomes obvious that strategies using s and s^* will display different volatilities, and different implementations of s^* will lead to different volatilities as well. In order to compare strategies, we will use mainly statistics that are normalized by volatility.

Obviously, many different ways to transform uncertainty into an alteration coefficient could be tested. Linear and kinked functions are probably the simplest. Among potential transformation functions, we could use concave or convex functions, like exponential or log functions.

5 Performance of Trend-Following strategies

We start by comparing the initial trend-following strategy with the alternative strategies calculating uncertainty over 20 days. We remind portfolios are rebalanced daily and that returns of all strategies are net of (estimated) transaction costs. To make it easier to compare absolute returns, the strategies are levered ex-post to a volatility of 10%. Table 1 and Figure 4 compare thirteen implementations. Outcomes are overall very similar, and we see this robustness as a positive sign, as the choice of the parameter(s) will not make a massive difference.

Interestingly, all specifications lead to an improved Sharpe ratio. Improvements are limited though, and given the MaxDrawdown (MDD) to vol ratio is deteriorated, we may wonder whether it is worth it. Looking at a broader set of results in Figures 5 and 6 could lead to a positive answer.

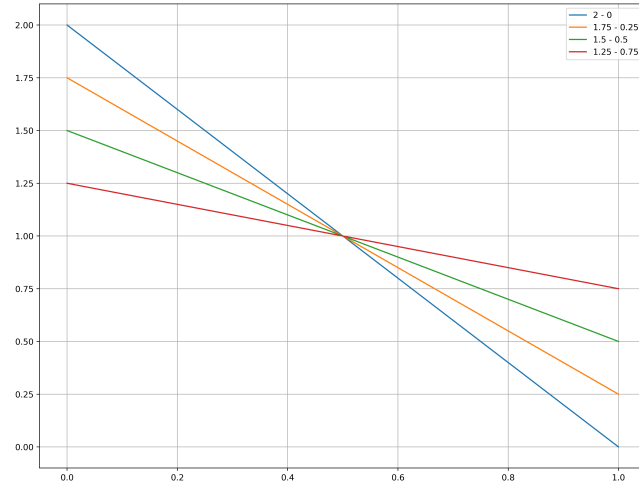
While the type of transformation used does not seem to make much of a difference at first glance, as displayed in Figure 5b, a closer look leads to the conclusion that the linear transformation offers a slight edge, as it tends to improve more the Sharpe ratio than other transformations for a given impact on the MDD to vol ratio, or, quite visible in the top left quadrant, lowers more the MDD to vol ratio for a given improvement in Sharpe ratio.

Rather than the transformation though, the lookback period appears to be the major factor, as displayed in 6a. Overall, short term lookback periods such as 10 or 15 days tend to improve the Sharpe ratio at the expense of the MDD to vol ratio. At the opposite of the spectrum, the 50-day lookback period is the only specification leading to lower Sharpe ratios and lower MDD to vol ratios. All outcomes using 30- and 40-day lookback periods save one are in the best quadrant, ie higher Sharpe ratios and lower MDD to vol ratios. As the lookback period used to calculate variability lengthens, there is a clear path from the top right quadrant to the bottom left quadrant.

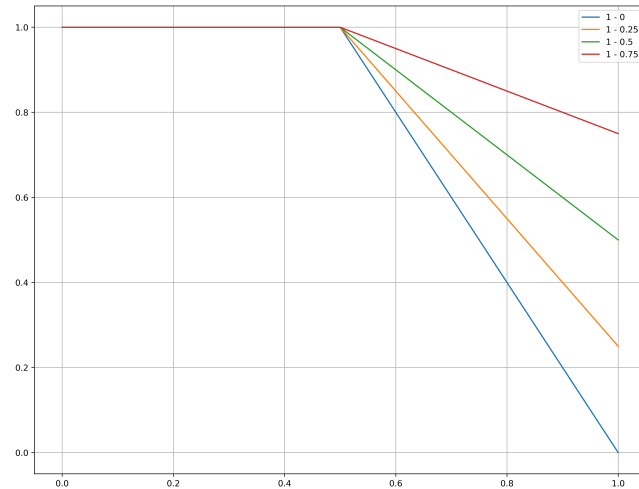
We then compute the excess return of alternative trend-following strategies vs the initial strategy, ie we go long the alternative strategy implementing and short the initial strategy, and test whether the expanding Sharpe ratio of this long-short strategy is different from 0 (for more details on the framework, see [6]). The expanding Sharpe ratios delivered by

Figure 3: Transformation function of uncertainty into alteration coefficient

(a) Linear transformation: low (resp. high) uncertainty is favored (resp. penalized)



(b) Kinked transformation: high uncertainty is penalized



(c) Kinked transformation: low uncertainty is favored

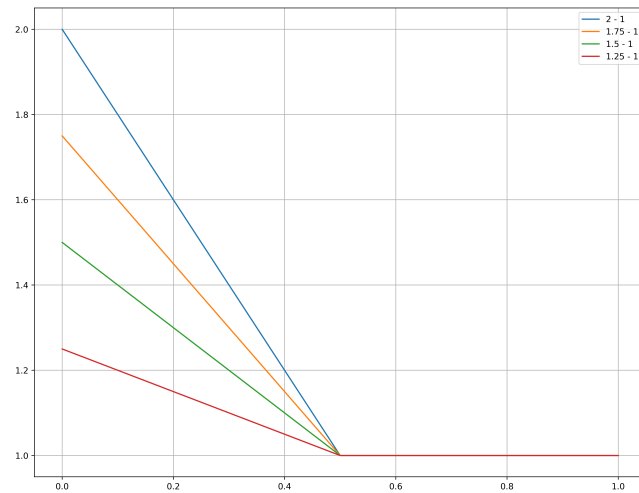
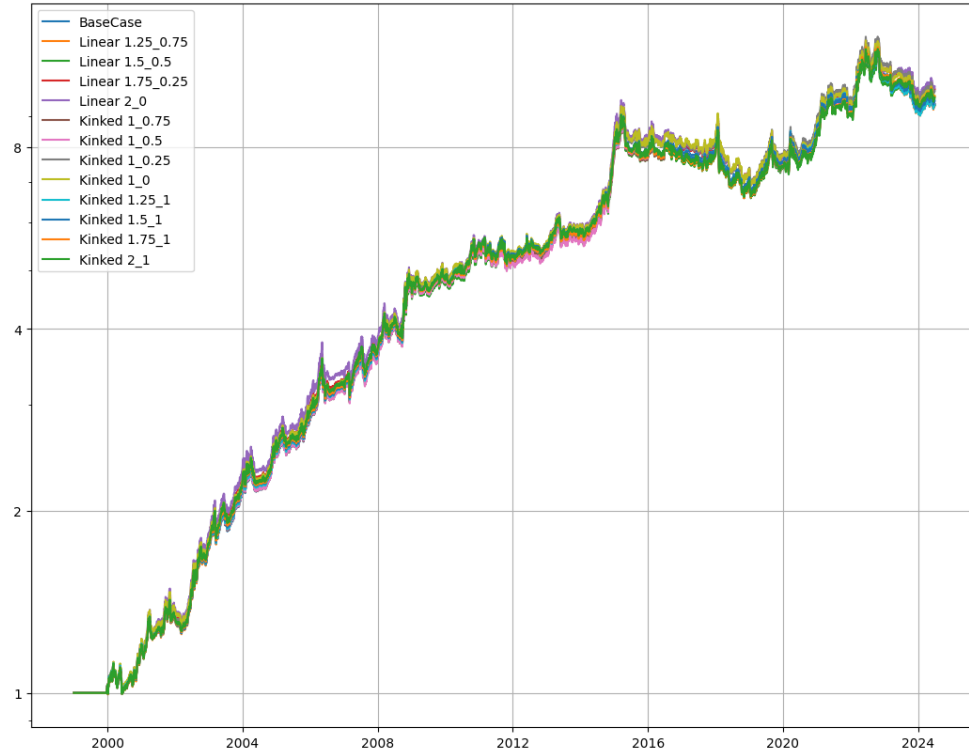


Figure 4: Cumulative returns of trend-following strategies without and with uncertainty adjustment using a 20-day lookback period, levered to the same realized volatility



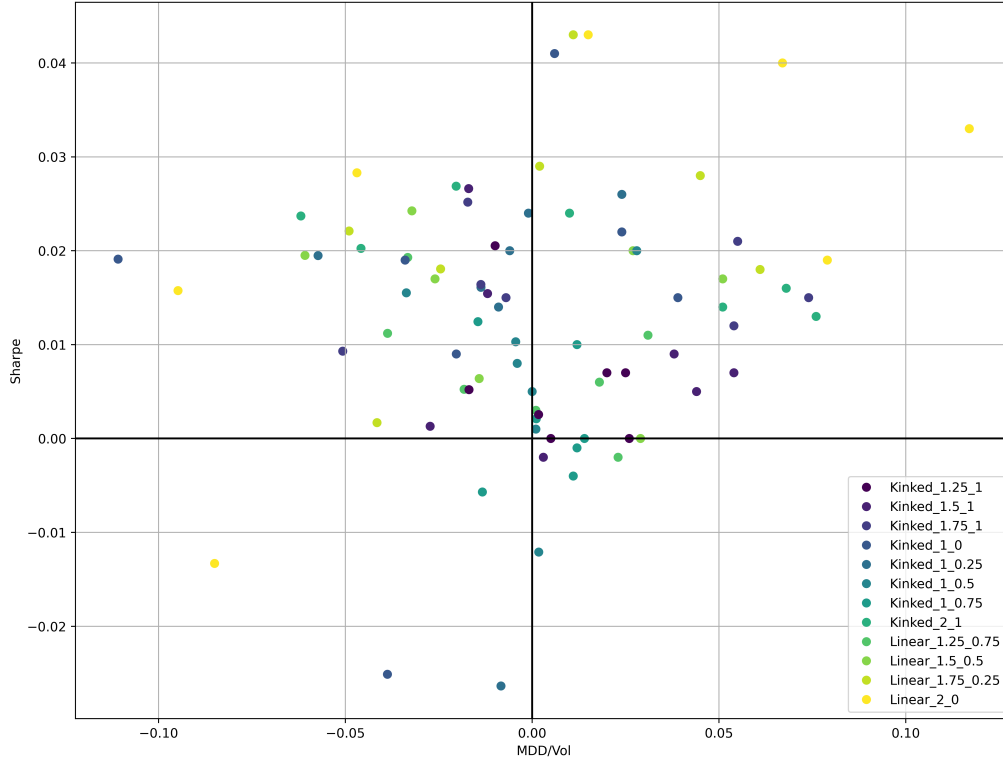
	BaseCase	Linear				Kinked right				Kinked left			
		± 0.25	± 0.5	± 0.75	± 1	1-0.75	1-0.5	1-0.25	1-0	1.25-1	1.5-1	1.75-1	2-1
Sharpe Ratio	0.92	0.92	0.92	0.95	0.95	0.92	0.93	0.95	0.94	0.92	0.93	0.94	0.94
MDD/Vol	2.5	2.6	2.6	2.7	2.8	2.6	2.5	2.5	2.6	2.6	2.7	2.7	2.7
Skewness	-0.4	-0.4	-0.4	-0.4	-0.5	-0.4	-0.4	-0.4	-0.4	-0.4	-0.4	-0.4	-0.4

Table 1: Key stats of trend-following strategies without and with uncertainty adjustment using a 20-day lookback period, levered to the same realized volatility

the long-short strategies (using 30- and 40- lookback periods) are displayed in Figure 7. Though p-values remain above 20%, the direction of travel for the p-values is overall down: as history grows, so do the odds the excess risk-adjusted returns delivered by the alternative strategies over the initial strategy are significantly different from 0.

Figure 5: Impact on Sharpe ratio and MDD/Vol ratio for all implementations

(a) Grouped by linear / kink parameters



(b) Grouped by kink type

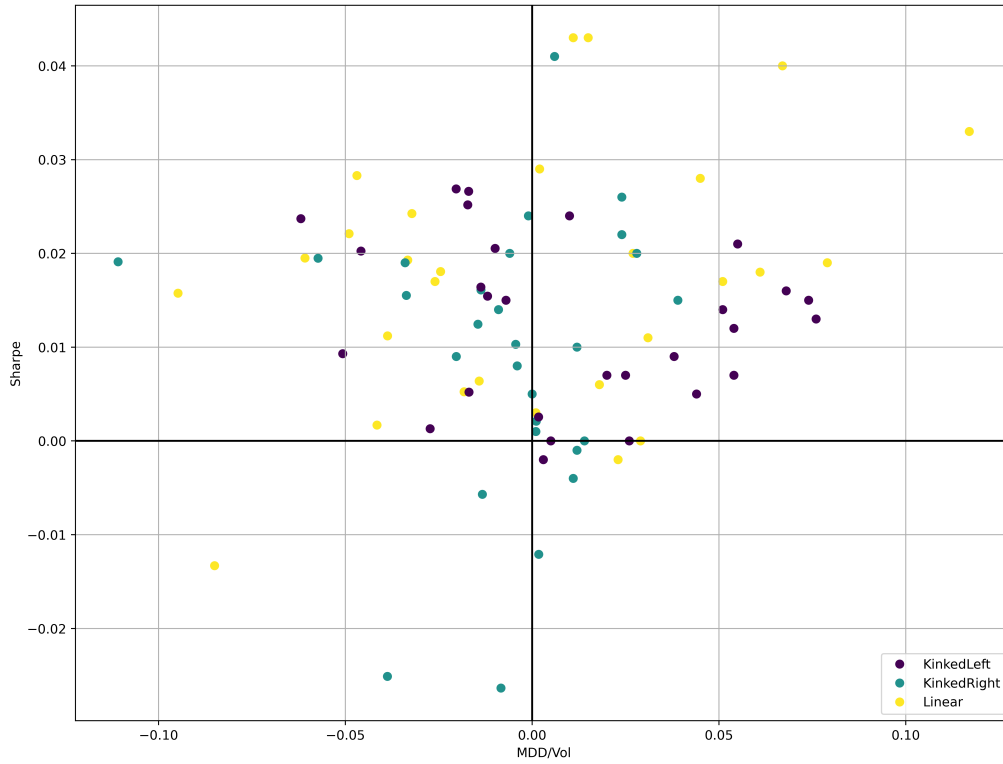
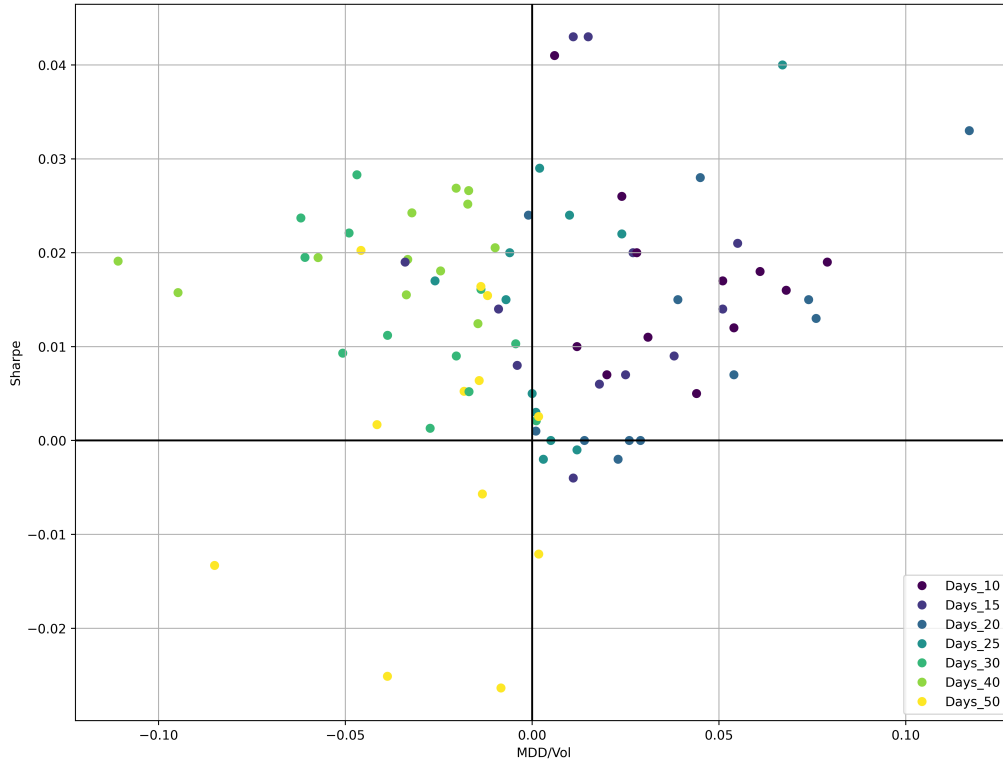


Figure 6: Impact on Sharpe ratio and MDD/Vol ratio for all implementations

(a) Grouped by lookback period



(b) Singling out 30- and 40-day lookback periods

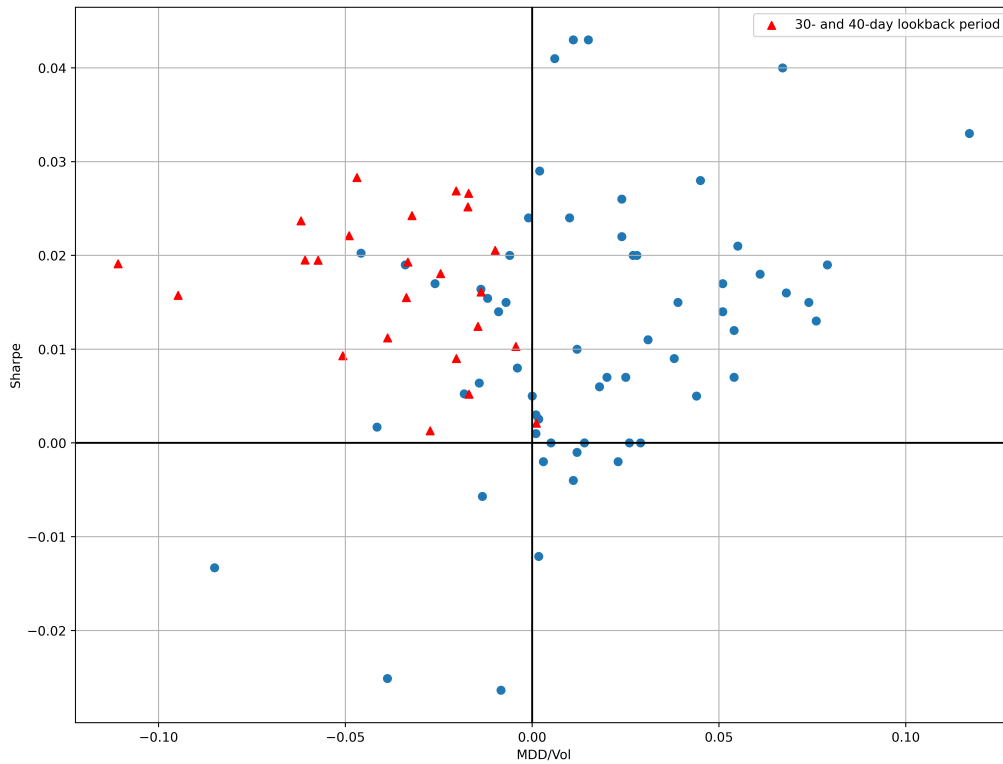
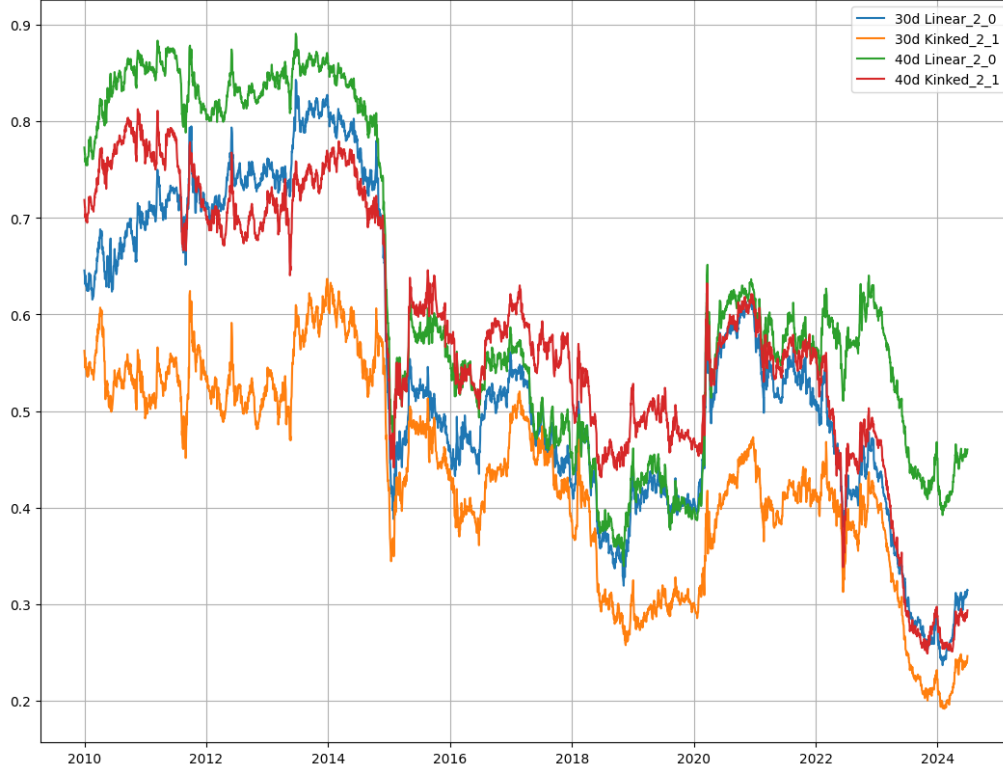


Figure 7: p-values of long-short portfolios, long (resp. short) a given alternative (resp. initial) trend-following strategy, for the test $H_0 : SR \leq 0$ vs $H_1 : SR > 0$. The figure shows the expanding p-values of the portfolios, i.e. the probability that H_0 is true.



6 Conclusion

Altering the trend-following signals according to their recent variability may add value, by improving the Sharpe ratio in most specifications tested.

The shape of the transformation may be impactful, however overall, the lookback period used to define variability seems to be the dominant factor, with a clear pattern. Short-term periods (up to one month) lead to improved Sharpe ratios at the expense of MDD to vol ratios, which also move higher. The sweet spot seems to be in between one and two months, as Sharpe ratios increase and MDD to vol ratios decrease. Over two months, both Sharpe and MDD to vol ratios tend to decrease.

It would be interesting to compare these results with alterations based on fractal analysis, using local estimates of the Hurst exponent, and eventually combine both views from fractal analysis and information theory, and adapt what was done in [4].

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